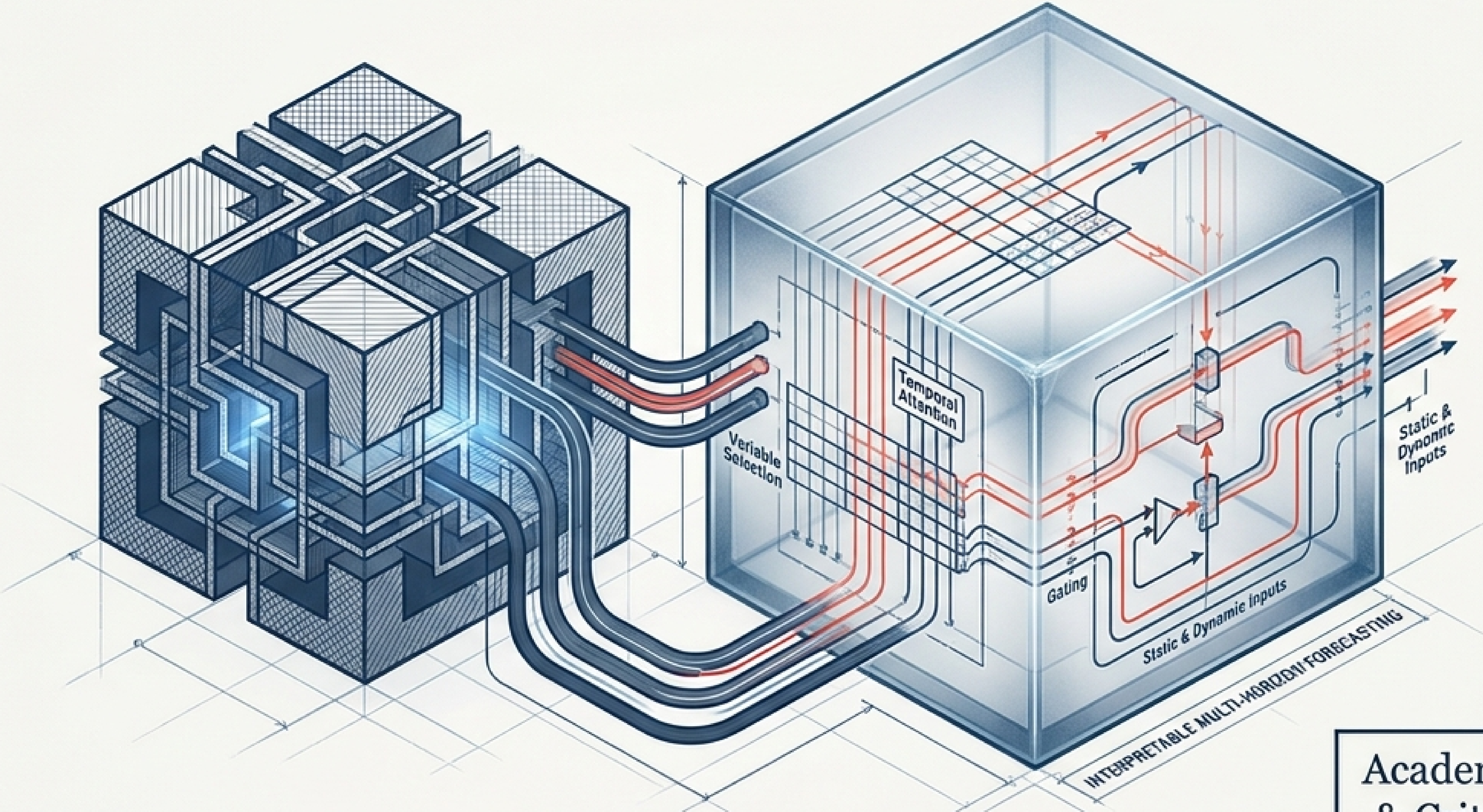
# Temporal Fusion Transformers (TFT)

## An Architectural Deconstruction of Interpretable Multi-Horizon Forecasting




Academic Deep-Dive
& Critical Synthesis

## Forecast Time (t)

Column 1 | Past | Future

### Static Metadata

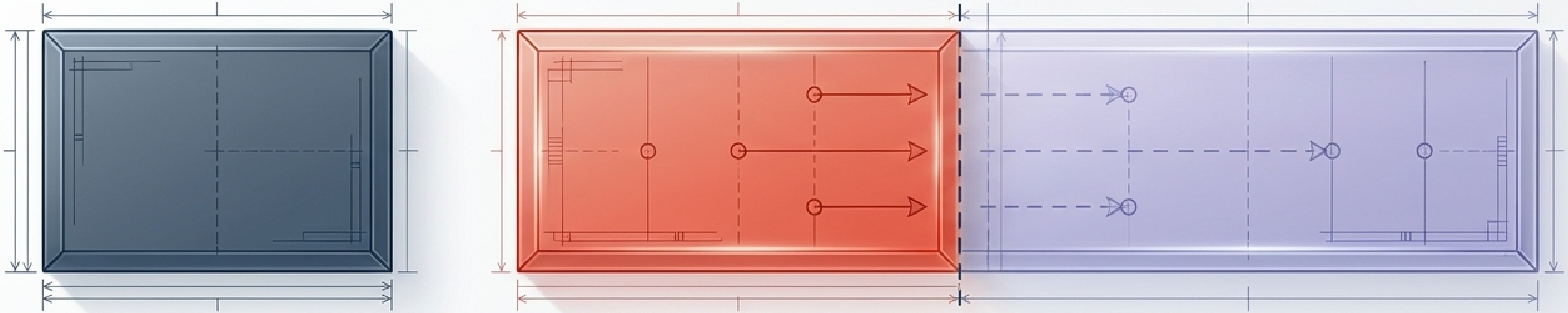
Time-invariant context (e.g., store location, patient genetic history).

### Observed Past Inputs

Time-varying data only known up to the present (e.g., historical foot traffic).

### Known Future Inputs

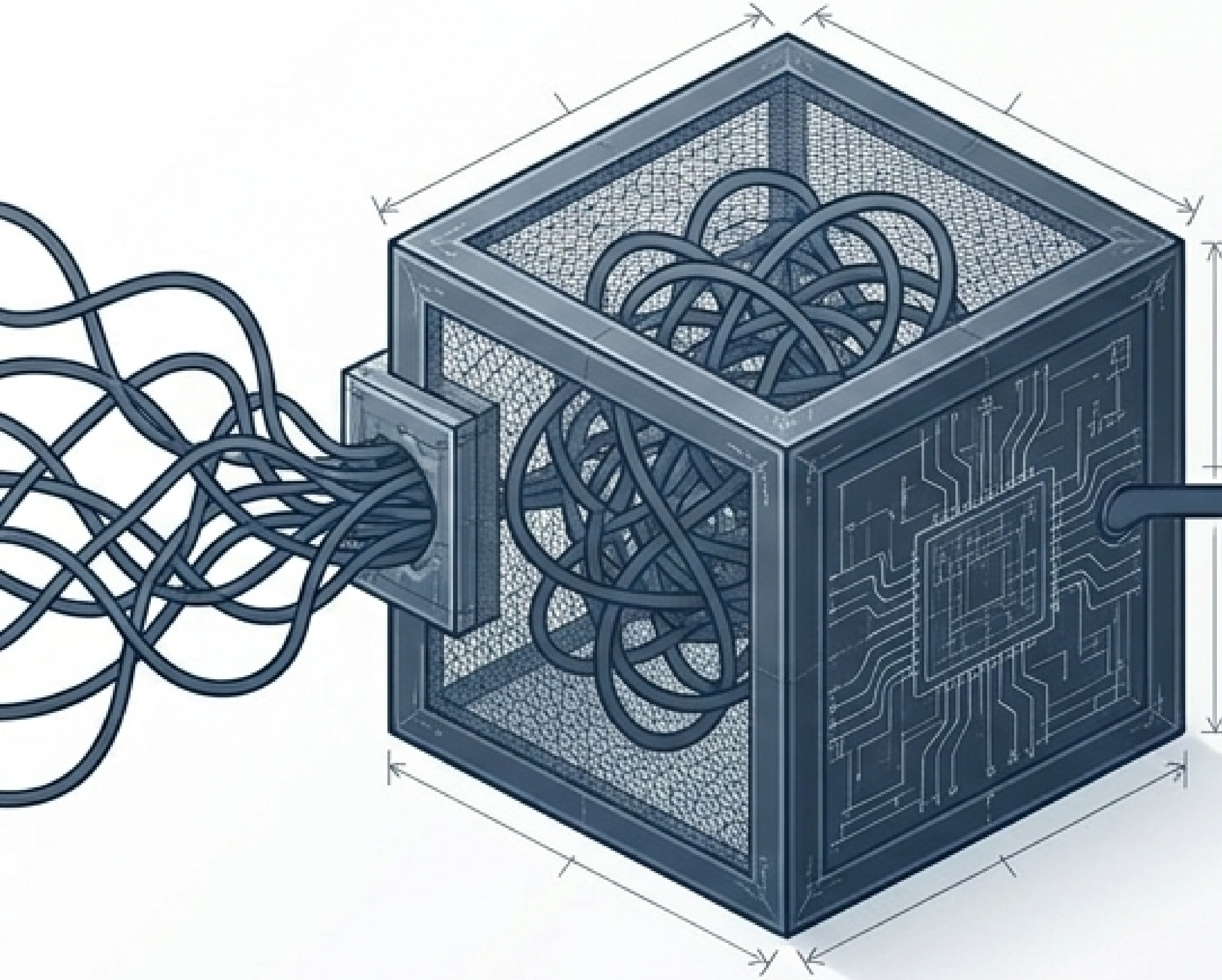
Time-varying data predetermined into the future (e.g., day-of-week, holidays).



**The Challenge:** Real-world forecasting requires blending these three highly heterogeneous data types without prior knowledge of how they interact with the target variable.

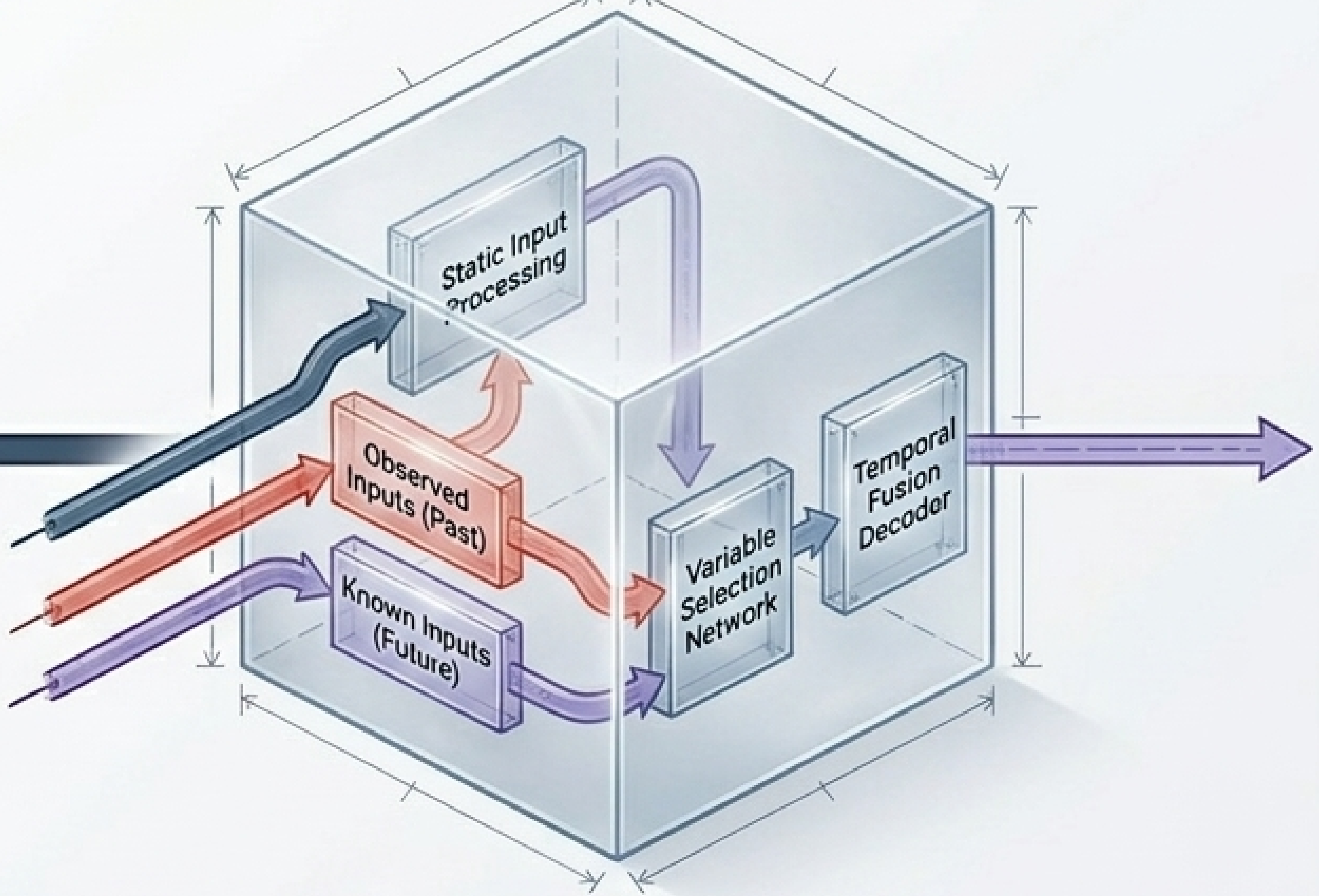
# The "Black-Box" Dilemma in Time Series

## The Status Quo: Standard Deep Learning



High accuracy, but relies on complex, non-linear parameter **interactions**. Post-hoc explainability (LIME, SHAP) fails because surrogate models do not natively respect the sequential time-ordering of inputs.

## The TFT Paradigm: Inherently Interpretable



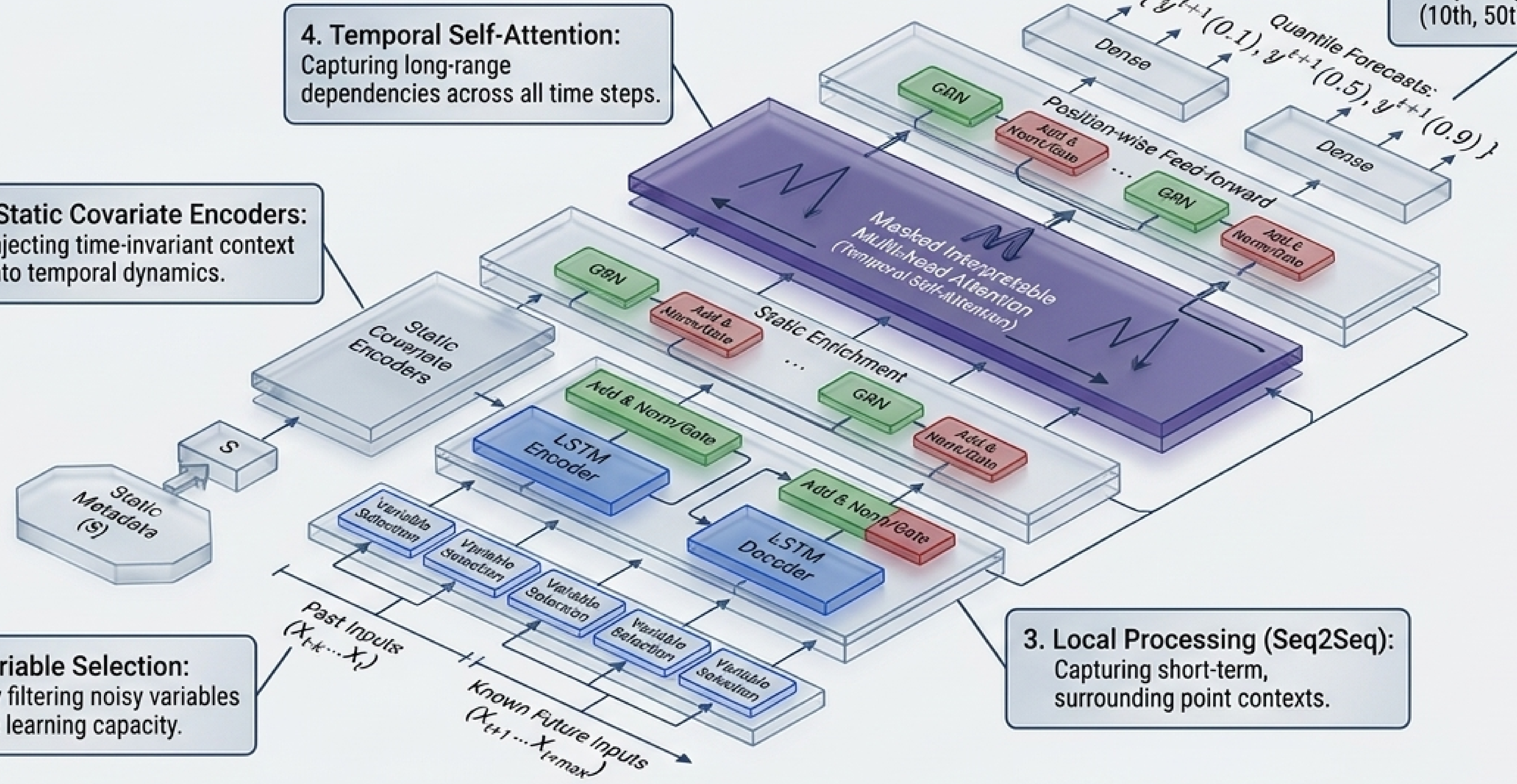

**TFT** uses specialized architectural components to route data. This allows for global extraction of variable importance, seasonal patterns, and regime shifts directly from the network's structural weights.

# Anatomy of the Temporal Fusion Transformer

**5. Quantile Forecasts:** Outputting prediction intervals (10th, 50th, 90th percentiles).

**4. Temporal Self-Attention:** Capturing long-range dependencies across all time steps.

**2. Static Covariate Encoders:** Injecting time-invariant context into temporal dynamics.

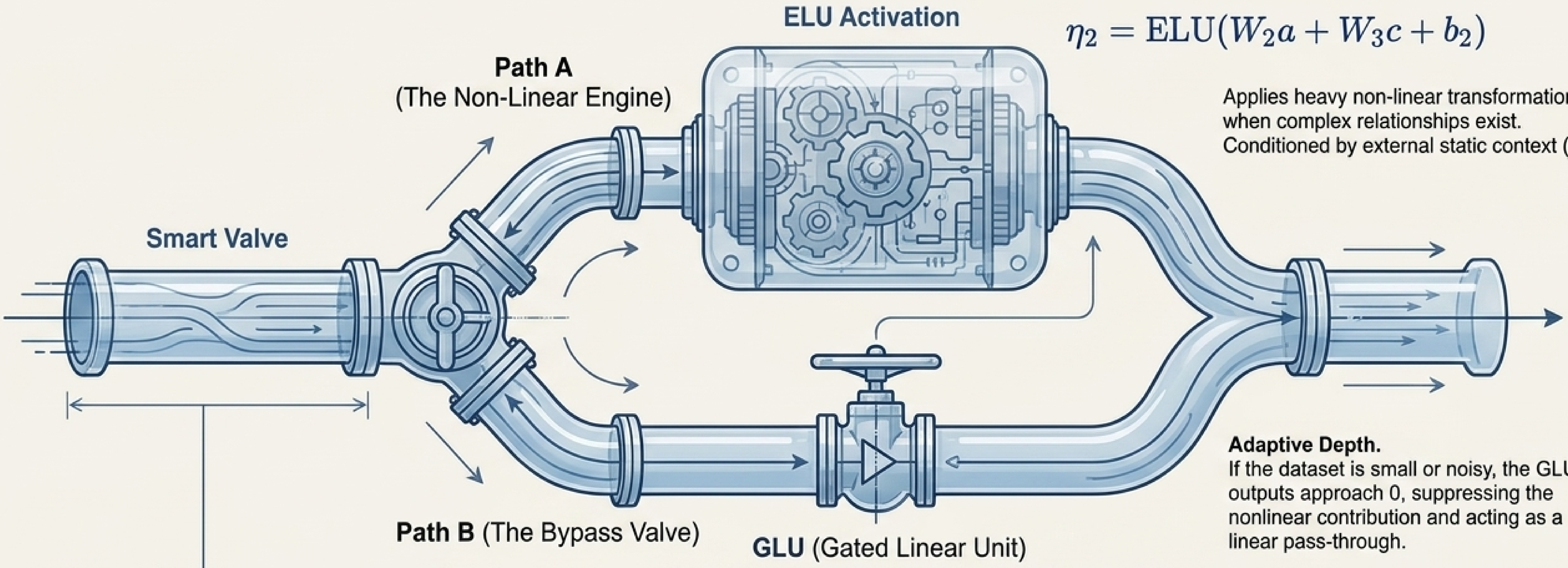



**1. Input Variable Selection:** Judiciously filtering noisy variables to preserve learning capacity.

**3. Local Processing (Seq2Seq):** Capturing short-term, surrounding point contexts.

**TFT abandons the monolithic network approach, instead deploying specialized canonical components to independently build feature representations.**

# Dynamic Routing: Gated Residual Networks (GRN)




$$\eta_2 = \mathrm{ELU}(W_2 a + W_3 c + b_2)$$

Applies heavy non-linear transformations when complex relationships exist. Conditioned by external static context (**c**).

**Adaptive Depth.**
If the dataset is small or noisy, the GLU outputs approach 0, suppressing the nonlinear contribution and acting as a linear pass-through.

**Strategic Purpose:** Prevents overfitting on small or volatile datasets by dynamically scaling architectural complexity.

# Instance-Wise Variable Selection

### Step 1: Categorical & Continuous Embeddings

Raw variables are transformed into $d_{\_\mathrm{model}}$-dimensional vectors.

### Step 2: Weight Generation

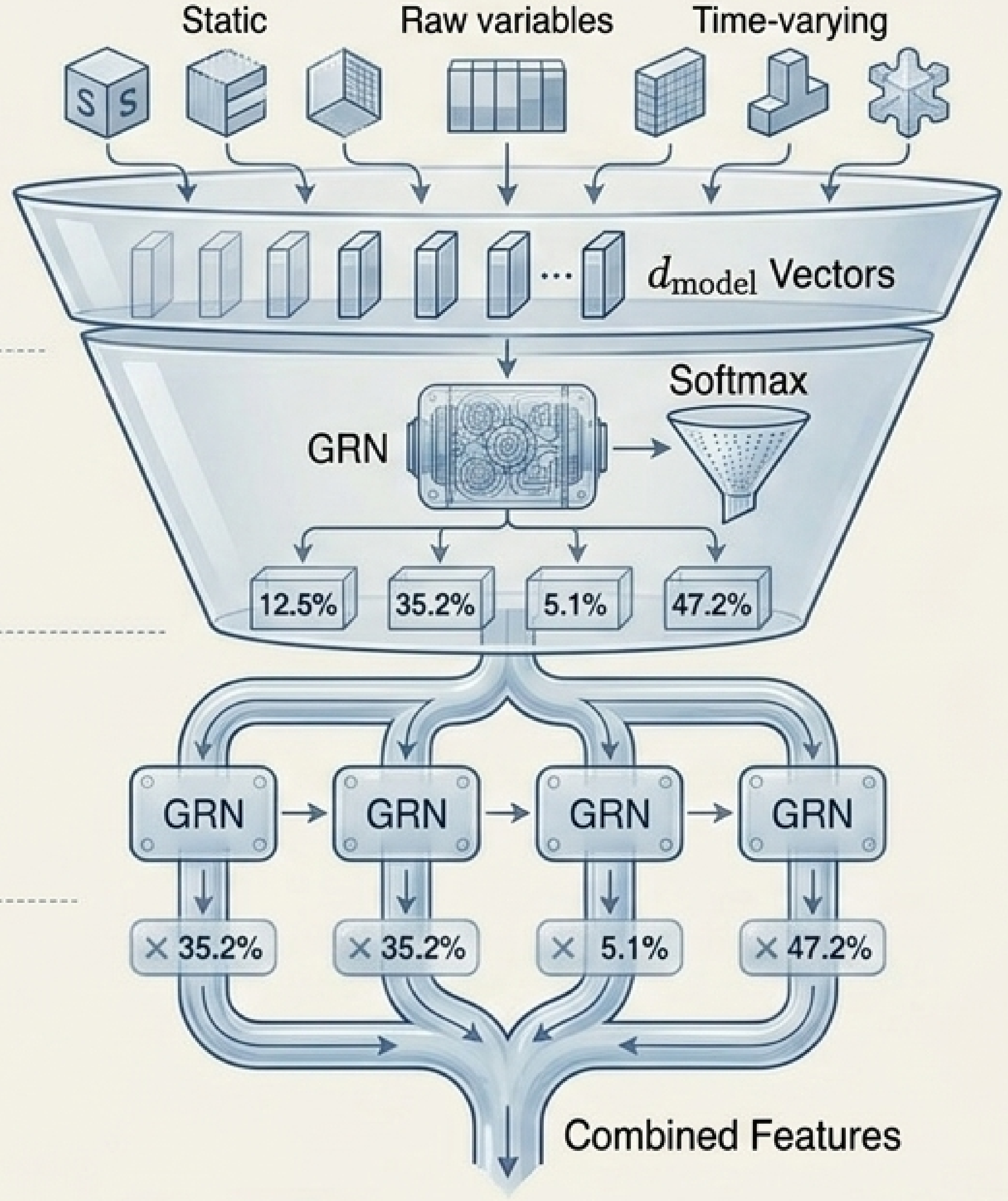
Flattened input vectors combined with static context generate variable selection weights ($v_{\chi t}$).

### Step 3: Independent Feature Processing

Each variable undergoes independent non-linear processing.

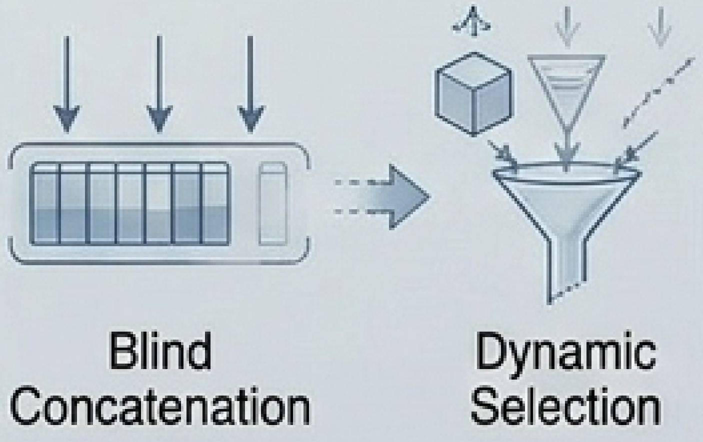
### Step 4: Weighted Combination

Processed features are multiplied by selection weights and combined.



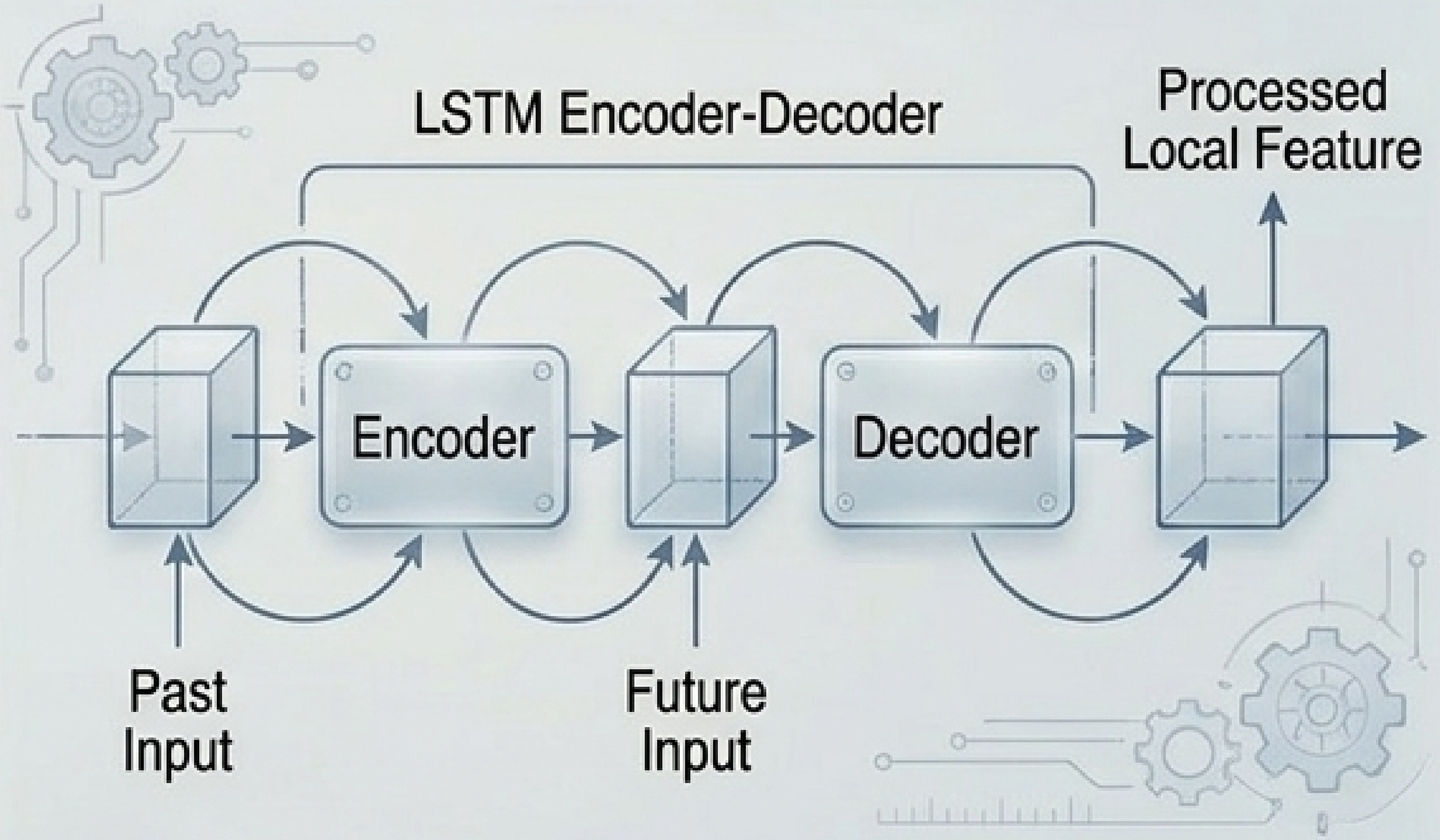

## Significance

Unlike standard models that blindly concatenate features, TFT dynamically discards non-predictive variables at each specific time step.



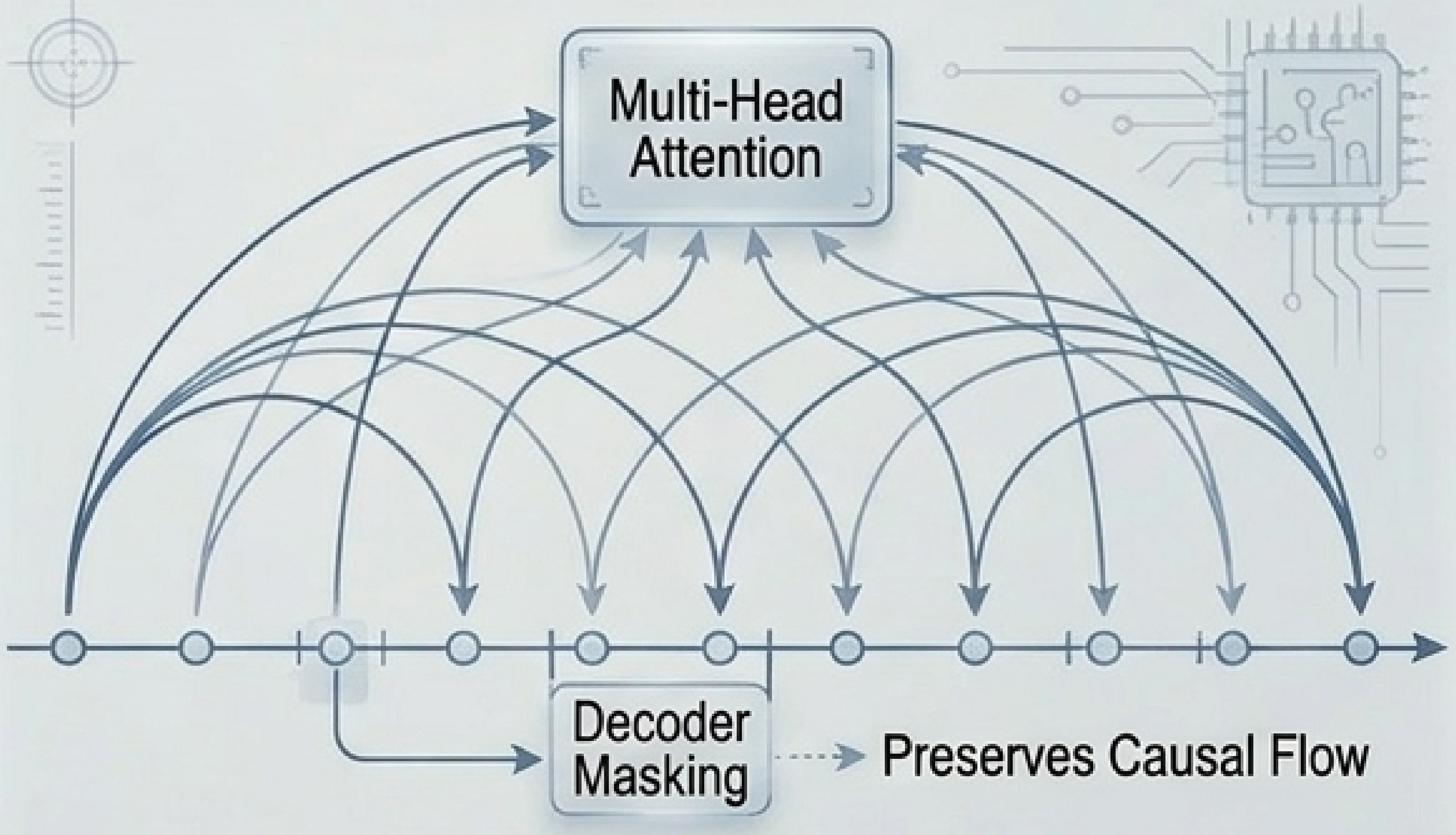
# Dual-Scale Temporal Processing

## **Local Context** (Seq2Seq Layer)




- Replaces standard positional encoding.
- Feeds past inputs into an encoder and known future inputs into a decoder.
- Identifies anomalies and change-points relying on immediate surrounding values.

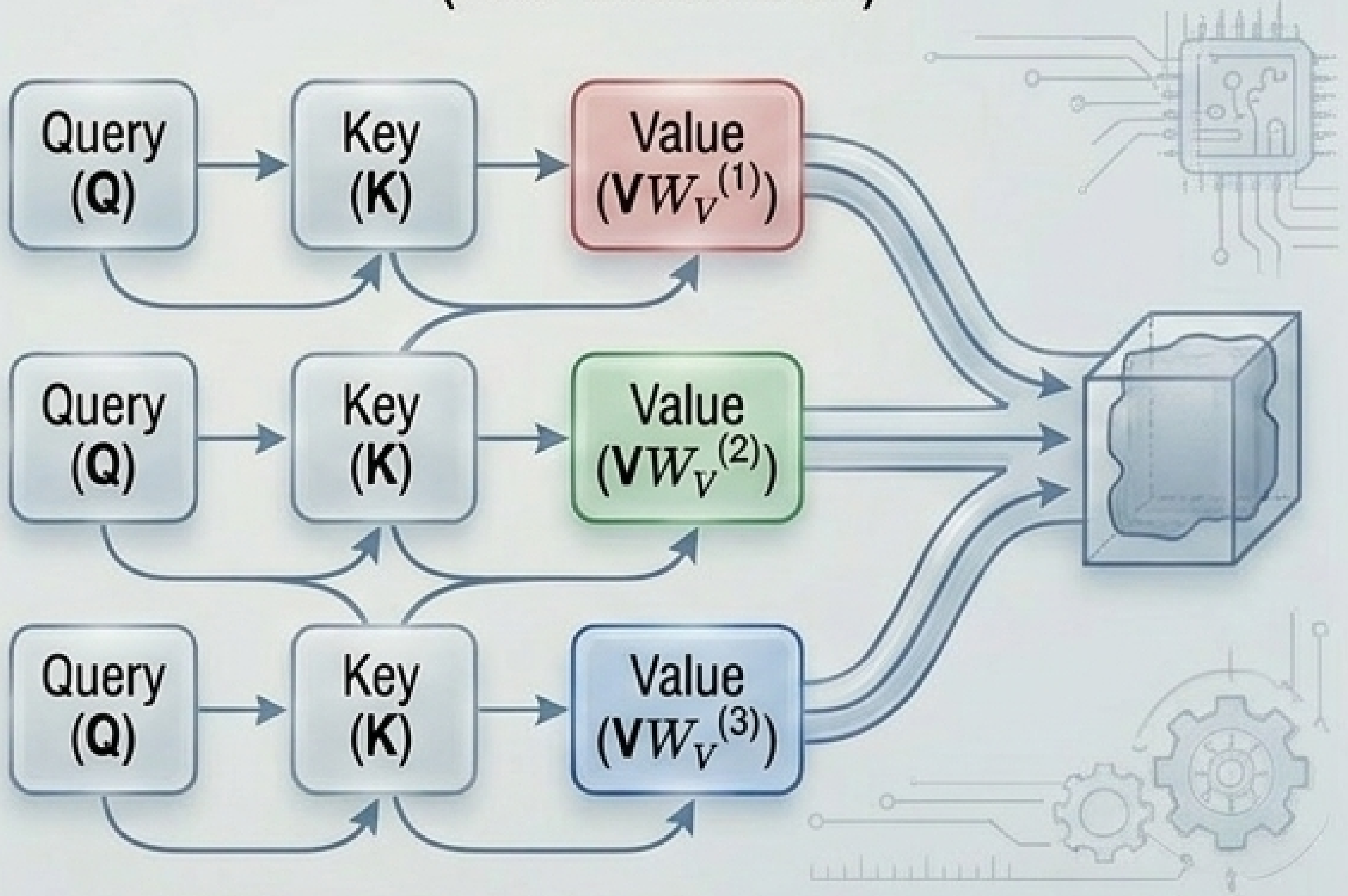
## **Global Context** (Self-Attention Layer)




- Applies multi-head attention to static-enriched temporal features.
- Uses decoder masking to strictly preserve causal flow.
- Captures long-range dependencies that recurrent architectures (LSTMs) mathematically fail to retain.

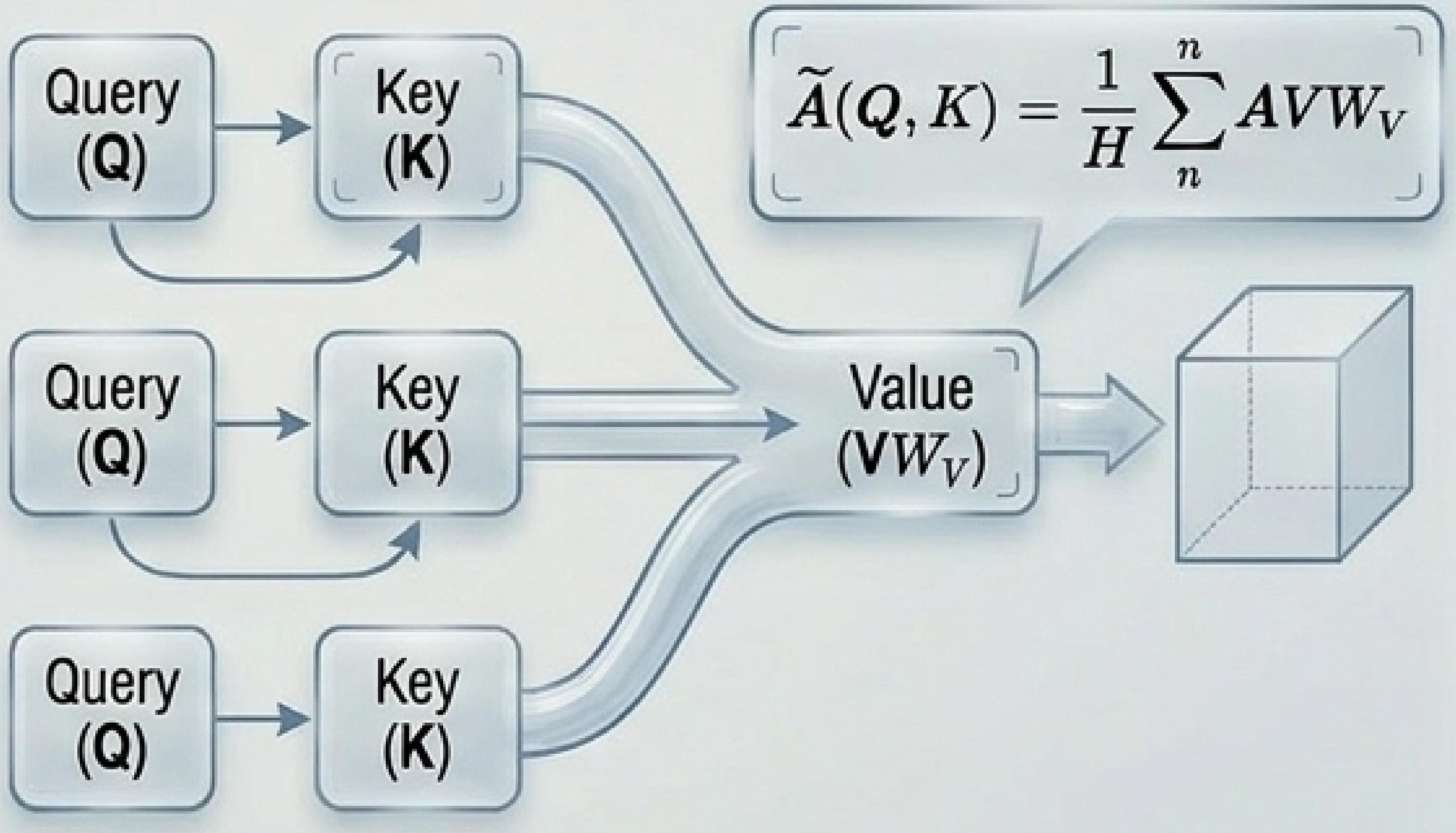
# Engineering Interpretable Attention

## Standard Multi-Head Attention
(The Black Box)




- Because every attention head uniquely transforms the values, the final weights cannot be traced back to original feature importance.

## TFT's Interpretable Multi-Head Attention
(The Glass Box)




- By sharing value weights across all heads, the heads act as a simple ensemble. This allows TFT to trace attention weights directly back to specific input features, yielding global interpretability.

# Empirical Validation: Performance Benchmarks

Electricity | Traffic | Retail | Volatility

Iterative Methods (Require future imputation):
DeepAR, DSSM, ConvTrans

Direct Methods:
Seq2Seq, MQRNN

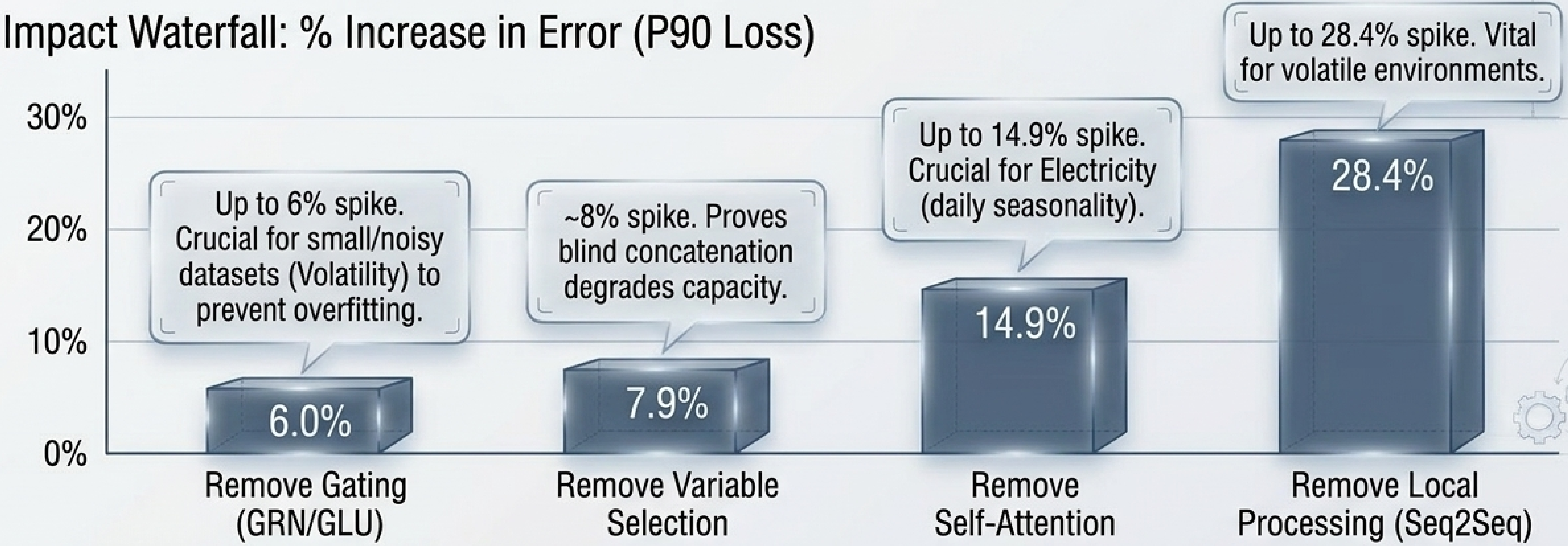
**+7% Improvement**

Average reduction in P50 (median) quantile loss vs. next best baseline.

**+9% Improvement**

Average reduction in P90 (worst-case risk) quantile loss vs. next best baseline.

# Ablation Synthesis: Proving the Architecture




**Conclusion:** The architecture contains no redundant parts; performance relies strictly on the mathematical synergy of localized and global components.

# Unlocking the Glass Box I: Global Variable Importance

## Static Covariates

- Item Number (0.230) ↑
- Store Number (0.161)
- Cluster (0.122)

The model automatically assigned the highest weights to entity identifiers without explicit instruction.

## Past Inputs

- Log Sales (0.324) ↑
- Month (0.122)
- Oil (0.081)

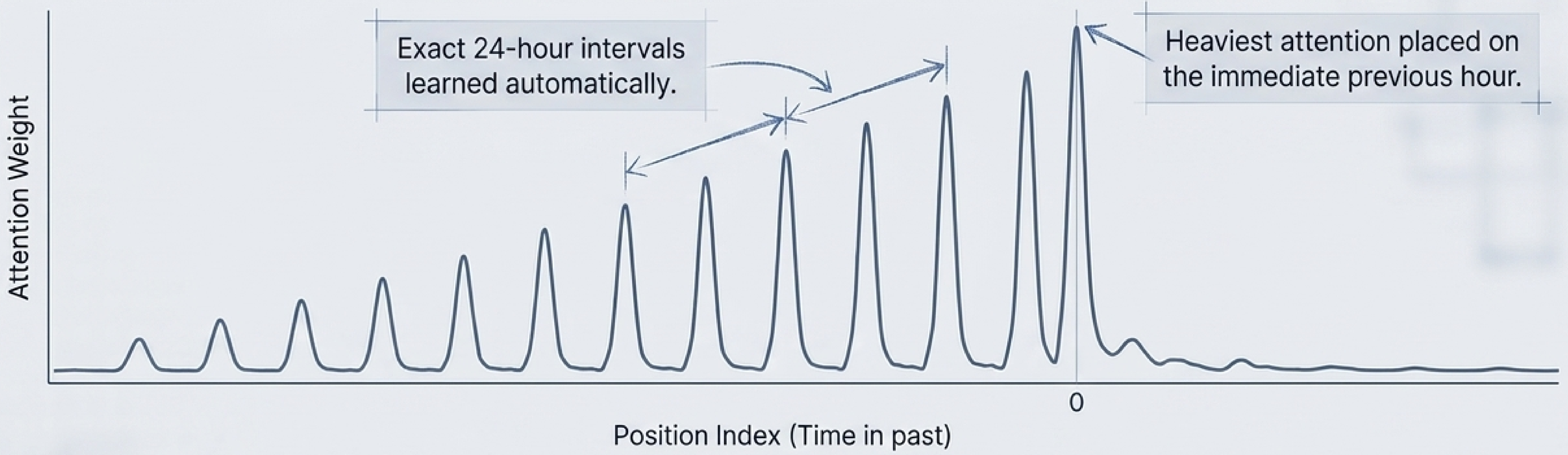
Past target data holds the vast majority of weight, confirming forecasts are primarily extrapolations of recent history.

## Future Inputs

- National Holidays (0.220)
- On-Promotion (0.170)
- Month (0.155)

The network correctly learned to heavily weight promotional and holiday data, mirroring human retail intuition.

# Unlocking the Glass Box II: Persistent Temporal Patterns



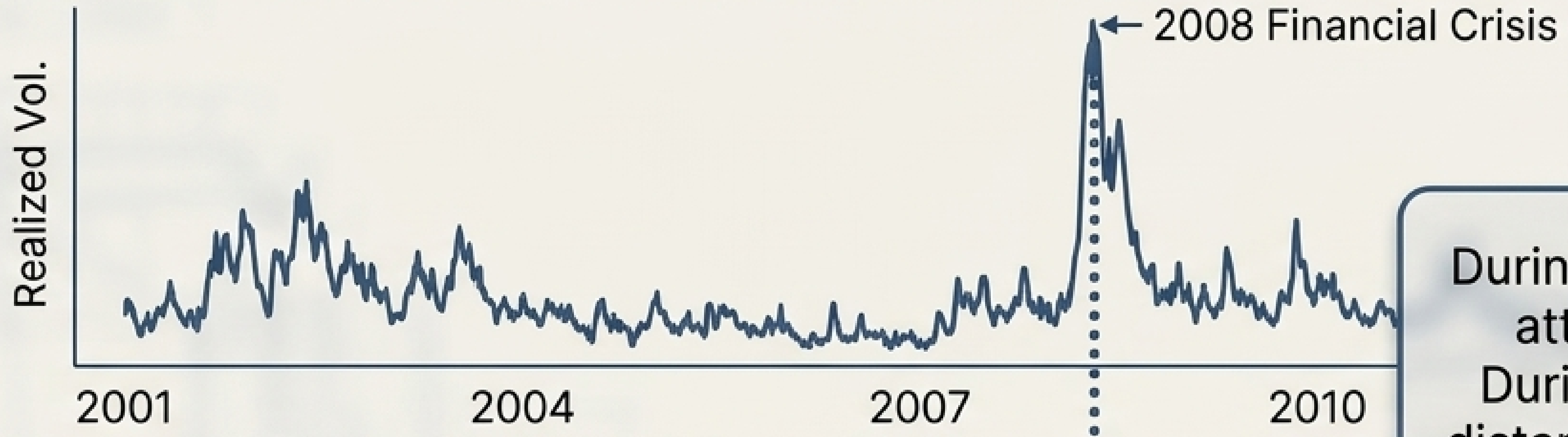

In traditional econometrics (ARIMA), seasonality must be manually hard-coded.

By analyzing the interpretable multi-head attention weights, TFT reveals exactly where it looks in the past to predict the future.

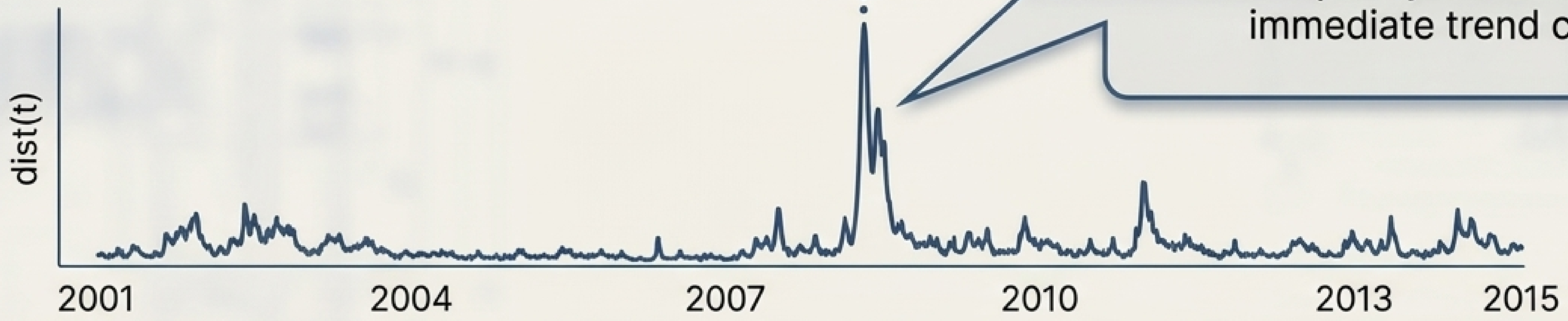
Electricity/Traffic spike at 24 hours. Retail spikes at 7 days. Volatility flattens entirely, acting as a smoothing moving-average filter.

# Unlocking the Glass Box III: Identifying Regime Shifts

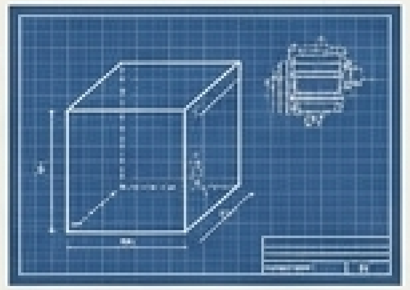
## The Event (S&P 500 Realized Volatility)




## The Model's Internal Logic (Bhattacharya Distance)




During stable periods, TFT distributes attention equally (low distance). During market crashes, the internal distance metric violently spikes as the model dynamically shifts its attention to focus purely on sharp, iom sharp, immediate trend changes.



Blueprint to Glass Box

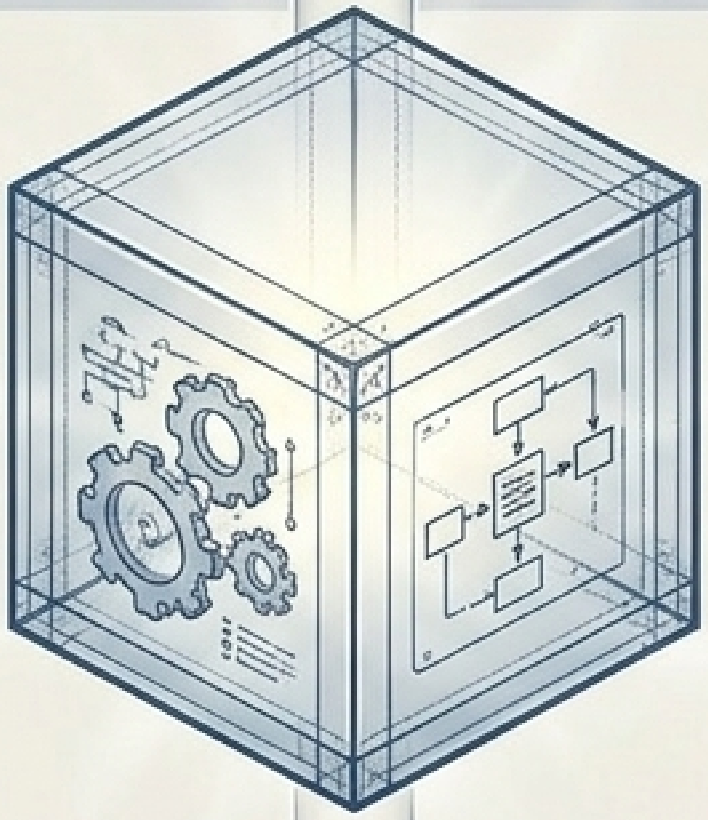
# Architectural Diagnostic Comparison

| | Iterative Auto-Regressive (DeepAR) | Direct Seq2Seq | Direct Transformer (ConvTrans) | Temporal Fusion Transformer (TFT) |
|---|---|---|---|---|
| Handles Heterogeneous Data? | Poor (Requires imputation) | Moderate | Moderate | Excellent (Dedicated encoders) |
| Captures Long-Range Dependencies? | Poor (Vanishing gradients) | Poor | Good | Excellent (Self-attention) |
| Prevents Overfitting on Noise? | Moderate | Moderate | Moderate | Excellent (GRN/GLU gating) |
| Global Interpretability? | None | None | Instance-specific only | Global (Shared-value attention) |

# Critical Synthesis: The TFT Balance Sheet

## Theoretical Elegance & Strengths

- **Unifies** heterogeneous data streams without requiring manual feature crossing.
- Establishes **State-of-the-Art** accuracy across diverse domains (Average 7-9% loss reductions).
- Provides actionable, global interpretability from internal weights, replacing flawed post-hoc explainers.



## Practical Disadvantages & Limitations

- **Computational Overhead**: Multi-head attention is memory intensive for highly extended sequences compared to simple RNNs.
- **Architectural Complexity**: Requires heavy hyperparameter tuning to prevent numerous GLUs from shutting down inappropriately.
- **Data Hunger**: Interpretability matrices are only statistically reliable when trained over massive, rich datasets.

**Final Verdict:** TFT transforms time series forecasting from a black-box estimation into an **interpretable science**—but at the cost of architectural and computational complexity.